

## Hồ Lam Sơn

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| BASIC INFORMATION      | <p>Name also written as: Lam-Son Ho, Son Lam Ho</p> <p>Current address: Hanoi, Vietnam</p> <p>Email: <a href="mailto:holamson2000@gmail.com">holamson2000@gmail.com</a> or <a href="mailto:son@urbanmetry.com">son@urbanmetry.com</a></p> <p>GitHub: <a href="https://github.com/sonlamho">github.com/sonlamho</a></p> <p>Website: <a href="https://sonlamho.github.io">https://sonlamho.github.io</a></p> <p>Languages: Vietnamese (native), English (fluent)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| PROFILE                | <p>Data scientist and Math PhD (University of Maryland) with nine years of experience leading machine learning projects in industry. Teaching experience in university courses in Statistics, Linear Algebra, Calculus, as well as Foundation of Machine Learning workshops at VietAI.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| EDUCATION              | <p><b>University of Maryland</b>, College Park, MD</p> <p>Ph.D., Mathematics, August 2014</p> <ul style="list-style-type: none"><li>• Advisor: Dr. William M. Goldman</li></ul> <p><b>Williams College</b>, Williamstown, MA</p> <p>B.A. with Honors, Mathematics and Computer Science (Double Major), June 2008</p> <ul style="list-style-type: none"><li>• Advisor: Dr. Thomas Garrity</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| PROFESSIONAL POSITIONS | <p><b>Urbanmetry Sdn Bhd</b>, Malaysia</p> <p>Lead Data Scientist, 09/2017 – 09/2026</p> <p>Designed, implemented, and refined all stages of several machine learning projects, including:</p> <ul style="list-style-type: none"><li>• Home price prediction: a model used by banks to assess loan applications and by our sister company UrbanNX in its investment decision process.</li><li>• Bus ridership prediction: a model used by major bus operators in Kuala Lumpur and Penang to plan changes to their bus networks.</li><li>• Internal models: A text classifier using Convolution Neural Nets, an image segmentation model that detects houses in satellite imagery of rural areas.</li></ul> <p>Other notable software projects:</p> <ul style="list-style-type: none"><li>• A high-performance exchange order book in Go for trading fractionalized properties.</li></ul> <p>Provided data science and technical training for selected team members.</p> <p><b>VietAI</b>, Vietnam</p> <p>Teaching Team Member (Hanoi), 2018 – 2019</p> <ul style="list-style-type: none"><li>• Taught sessions for “Foundations of Machine Learning” course at VietAI, a non-profit organization building a community of AI practitioners and experts in Vietnam. Course topics: Intro to Neural Networks, Back-propagation algorithm and modifications, loss functions, regularization methods.</li></ul> <p><b>Université de Sherbrooke</b>, Quebec, Canada</p> <p>Postdoctoral Researcher, Mathematics, 08/2015 – 08/2017</p> <ul style="list-style-type: none"><li>• Advisor: Dr. Virginie Charette</li></ul> |

**University of Luxembourg**, Luxembourg

Postdoctoral Researcher, Mathematics, 09/2014 – 03/2015

- Advisor: Dr. Jean-Marc Schlenker

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| UNIVERSITY<br>TEACHING<br>EXPERIENCE | <p><b>Lecturer</b> – Planned lessons and assignments, lectured 3 hours per week per section (5 hours per week in summer terms), held office hours, and graded coursework.</p> <p>STAT 100 (at UMD) – Elementary Statistics and Probability      Spring 2014</p> <p>MATH 221 (at UMD) – Elementary Calculus II      Summer 2013</p> <p>MATH 461 (at UMD) – Linear Algebra for Scientists and Engineers      Summer 2011</p> <p>STAT 100 (at UMD) – Elementary Statistics and Probability      Fall 2009</p> <p><b>Teaching Assistant</b> – Helped plan assignments, taught or led discussion sessions 1.5 hours per week, and graded coursework.</p> <p>Analyse 1 (in Luxembourg) – Real Analysis 1      Fall 2014</p> <p>MATH 140 (at UMD) – Calculus I      Fall 2010</p> <p>MATH 220 (at UMD) – Elementary Calculus I      Spring 2010</p> <p><b>Grader</b> – Graded coursework and answered students' questions.</p> <p>MATH 432 (at UMD) – Introduction to Topology      Fall 2012</p> <p>MATH 431 (at UMD) – Geometry for Computer Graphics      Fall 2012</p> <p>MATH 341 (at UMD) – Linear Algebra and Differential Equations      Spring 2012</p> |
| TECHNICAL SKILLS                     | <ul style="list-style-type: none"><li>• Data science: Python (NumPy, pandas, scikit-learn, SciPy, XGBoost, Tensorflow) and SQL (PostgreSQL)</li><li>• Other languages and frameworks: Go, Ruby on Rails, Clojure</li><li>• AI-assisted software development with Claude Code</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| RESEARCH &<br>PUBLICATIONS           | <ul style="list-style-type: none"><li>• Chapter 8 of <i>Eighteen Essays in Non-Euclidean Geometry</i>. EMS IRMA Lectures in Mathematics and Theoretical Physics, 2019.</li><li>• “Deformations of fundamental group representations and earthquakes on <math>SO(n, 1)</math> surface groups.” <i>Geometriae Dedicata</i>, 2017. doi:10.1007/s10711-017-0226-0</li><li>• “On conformally flat circle bundles over surfaces.” Ph.D. thesis, University of Maryland, 2014. Preprint: <a href="https://arxiv.org/abs/1412.5824">arXiv:1412.5824</a></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| REFERENCES                           | <p>Available on request.</p> <p>(Updated: 2 October 2026)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |